



ANSWER KEY

1. (c) 2. (a) 3. (a) 4. (a) 5. (b) 6. (d) 7. (b) 8. (a) 9. (c) 10. (d)
 11. (c) 12. (c) 13. (b) 14. (b) 15. (b) 16. (c) 17. (b) 18. (a) 19. (d) 20. (d)
 21. [29] 22. [171] 23. [1] 24. [84] 25. [210] 26. [315] 27. [8] 28. [51] 29. [102] 30. [49]

Hints and Solutions

1. (c) $S_1 = (1999 + 24)^{2022} - (1999)^{2022}$

$$\Rightarrow {}^{2022}C_1 (1999)^{2021} (24) + {}^{2022}C_2 (1999)^{2020} (24)^2 + \dots \text{so on}$$

S_1 is divisible by 8. Hence, S_1 is correct.

$$S_2 : 13(13^n) - 11n - 13$$

$$13^n = (1 + 12)^n = 1 + 12n + {}^nC_2 12^2 + {}^nC_3 12^3 \dots$$

$$13(13^n) - 11n - 13 = 145n + {}^{13n}C_2 12^2 + {}^{13n}C_3 12^3 \dots$$

If $n = 144m, m \in \mathbb{N}$, then it is divisible by 144

For infinit value of n .

2. (a) We have $\left(2x^2 + \frac{1}{2x}\right)^{11}$

$$T_{r+1} = {}^{11}C_r (2x^2)^{11-r} \left(\frac{1}{2x}\right)^r$$

$$= {}^{11}C_r 2^{11-2r} x^{22-3r}$$

for coefficient of x^{10}

$$22 - 3r = 7 \Rightarrow r = 5$$

for coefficient of x^7

$$22 - 3r = 7 \Rightarrow r = 5$$

$$\therefore \text{Coefficient of } x^{10} = {}^{11}C_4 \cdot 2^3$$

$$\text{Coefficient of } x^7 = {}^{11}C_5 \cdot 2^1$$

$\therefore$ Absolute difference

$$= {}^{11}C_4 \cdot 2^3 - {}^{11}C_5 \cdot 2^1$$

$$= \frac{11 \times 10 \times 9 \times 8}{120} \times 8 - \frac{11 \times 10 \times 9 \times 8 \times 7}{120} \times 2$$

$$= 11 \times 10 \times 3 \times 8 - 11 \times 3 \times 4 \times 7$$

$$= 11 \times 3 \times 4 \times (20 - 7)$$

$$= 11 \times 12 \times 13$$

$$= 12 \times (12 - 1) \times (12 + 1)$$

$$= 12(12^2 - 1)$$

$$= 12^3 - 12$$

3. (a) $x = (8\sqrt{3} + 13)^{13} = {}^{13}C_0 \cdot (8\sqrt{3})^{13} + {}^{13}C_1 (8\sqrt{3})^{12} (13)^1 + \dots$

$$\text{Let } x' = (8\sqrt{3} - 13)^{13} = {}^{13}C_0 (8\sqrt{3})^{13} - {}^{13}C_1 (8\sqrt{3})^{12} (13)^1 + \dots$$

$$x - x' = 2 \left[{}^{13}C_1 \cdot (8\sqrt{3})^{12} (13)^1 + {}^{13}C_3 (8\sqrt{3})^{10} \cdot (13)^3 \dots \right]$$

Therefore, $x - x'$ is an even

integer, hence $[x]$ is even Now,

$$y = (7\sqrt{2} + 9)^9 = {}^9C_0 (7\sqrt{2})^9 + {}^9C_1 (7\sqrt{2})^8 (9)^1 + {}^9C_2 (7\sqrt{2})^7 (9)^2 \dots$$

$$y' = (7\sqrt{2} - 9)^9 = {}^9C_0 (7\sqrt{2})^9 - {}^9C_1 (7\sqrt{2})^8 (9)^1 + {}^9C_2 (7\sqrt{2})^7 (9)^2 \dots$$

$$y - y' = 2 \left[{}^9C_1 (7\sqrt{2})^8 (9)^1 + {}^9C_3 (7\sqrt{2})^6 (9)^3 + \dots \right]$$

$y - y' = \text{Even integer}$, hence $[y]$ is even

4. (a) $25^{190} - 19^{190} - 8^{190} + 2^{190}$

$$= 25^{190} - 8^{190} + 2^{190} - 19^{190}$$

$a^{2m} - b^{2m}$ is divisible by $(a - b)$

So, whole expression is divisible by 17

And $25^{190} - 19^{190} = \text{Even}$

$$2^{190} - 8^{190} = \text{Even}$$

Hence, expression is also divisible by 2

Hence, expression is divisible by $17 \times 2 = 34$

5. (b) $\left(ax^2 + \frac{1}{2bx}\right)^{11}$

$$\begin{aligned} T_{r+1} &= {}^{11}C_r (ax^2)^{11-r} \cdot \left(\frac{1}{2bx}\right)^r \\ &= {}^{11}C_r a^{11-r} \cdot \left(\frac{1}{2b}\right)^r \cdot x^{22-2r-r} = {}^{11}C_r a^{11-r} \cdot \left(\frac{1}{2b}\right)^r \cdot x^{22-3r} \\ \therefore 22-3r &= 7 \\ \Rightarrow 3r &= 15 \\ \Rightarrow r &= 5 \end{aligned}$$

Coefficient of $x^7 = \frac{{}^{11}C_5 a^6}{32b^5}$

Again, $\left(ax - \frac{1}{3bx^2}\right)^{11}$

$$\begin{aligned} T_{r+1} &= {}^{11}C_r (ax)^{11-r} \left(-\frac{1}{3bx^2}\right)^r \\ &= {}^{11}C_r a^{11-r} \cdot \left(\frac{-1}{3b}\right)^r \cdot x^{11-r-2r} \\ \therefore 11-3r &= -7 \\ \Rightarrow 3r &= 18 \\ \Rightarrow r &= 6 \end{aligned}$$

Coefficient of $x^{-7} = \frac{{}^{11}C_6 a^5}{3^6 b^6}$

According to the question, $\frac{{}^{11}C_5 a^6}{32b^5} = \frac{{}^{11}C_6 a^5}{3^6 b^6}$

$$\Rightarrow 729ab = 32$$

6. (d) $\frac{{}^nC_{r-1}}{1} = \frac{{}^nC_r}{5} = \frac{{}^nC_{r+1}}{20}$

$$\therefore \frac{{}^nC_r}{{}^nC_{r-1}} = 5 \Rightarrow \frac{\frac{n}{n-r+1} \cdot \frac{n-r+1}{r}}{\frac{n}{n-r+1} \cdot \frac{n-r+1}{r-1}} = 5$$

$$\Rightarrow \frac{n-r+1}{r} = 5 \Rightarrow n = 6r - 1 \quad \dots(i)$$

$$\therefore \frac{{}^nC_{r+1}}{{}^nC_r} = 4 \Rightarrow \frac{n-r}{r+1} = 4 \Rightarrow n = 5r + 4 \quad \dots(ii)$$

from (i) and (ii), $r = 5$, $n = 29$

Coefficient of fourth term = ${}^{29}C_3 = 3654$

7. (b) $\left(1-x^2+3x^3\right)\left(\frac{5}{2}x^3-\frac{1}{5x^2}\right)^{11}, x \neq 0$

$(r+1)$ th term for expansion of $\left(\frac{5}{2}x^3-\frac{1}{5x^2}\right)^{11}$

$${}^{11}C_r \times \left(\frac{5}{2}(x^3)\right)^{11-r} \times \left(-\frac{1}{5}x^{-2}\right)^r$$

$${}^{11}C_r \left(\frac{5}{2}\right)^{11-r} \left(-\frac{1}{5}\right)^r x^{33-3r-2r}$$

Coeff of x^0 in given expression

$$\therefore {}^{11}C_r \left(\frac{5}{2}\right)^{11-r} \left(-\frac{1}{5}\right)^r x^{35-5r}$$

So, $r = 7$

$$\begin{aligned} \therefore \text{The independent of } x &= -{}^{11}C_7 \left(\frac{5}{2}\right)^{11-7} \left(-\frac{1}{5}\right)^7 \\ &= 330 \times \frac{5^4}{2^4} \times \frac{1}{5^7} = \frac{33}{200} \end{aligned}$$

$\therefore$ Coefficient of term independent of x .

8. (a) $(2021)^{2022} + (2022)^{2021} = (2023-2)^{2022} + (2023-1)^{2021}$

$$\begin{aligned} &= 7k_1 + 2^{2022} + (2023-1)^{2021} \\ &= 7k_1 + 2^{2022} + 7k_2 - 1 \\ &= 7(k_1 + k_2) + 8^{674} - 1 \\ &= 7(k_1 + k_2) + 7k_2 + 1 - 1 \\ &= 7(k_1 + k_2 + k_2) \end{aligned}$$

So, the given number is divisible by 7.

Hence, the remainder is zero.

9. (c) Given, $(1-y)^m (1+y)^n$

The coefficient of $y (a_1) = 1 \cdot {}^nC_1 + {}^mC_1 (-1) = n - m$

$\dots(i)$

coefficient of $y^2 (a_2) = 1 \cdot {}^nC_2 - {}^mC_1 \cdot {}^nC_1 + 1 \cdot {}^mC_2 = 10$

$\dots(ii)$

From eqn (i) and equ (ii)

$$\frac{n(n-1)}{2} - mn + \frac{m(m-1)}{2} = 10$$

$$\Rightarrow m^2 + n^2 - 2mn - (n+m) = 20$$

$$\Rightarrow (n-m)^2 - (n+m) = 20 \Rightarrow n+m = 80$$

10. (d) $S_1 = -{}^{15}C_1 + 2 \cdot {}^{15}C_2 - \dots - {}^{15}C_{15}$

$$\begin{aligned} &= \sum_{r=1}^{15} (-1)^r \cdot r \cdot {}^{15}C_r = 15 \sum_{r=1}^{15} (-1)^{r-1} {}^{14}C_{r-1} \\ &= 15 (-{}^{14}C_0 + {}^{14}C_1 - \dots - {}^{14}C_{14}) = 15(0) = 0 \\ S_2 &= {}^{14}C_1 + {}^{14}C_3 + \dots + {}^{14}C_{11} \\ &= ({}^{14}C_1 + {}^{14}C_3 + \dots + {}^{14}C_{11} + {}^{14}C_{13}) - {}^{14}C_{13} \\ &= 2^{13} - 14 \\ \therefore S_1 + S_2 &= 2^{13} - 14 \end{aligned}$$

11. (c) General term of

$$(\alpha x^{1/9} + \beta x^{-1/6})^{10} = {}^{10}C_r (\alpha x^{1/9})^{10-r} (\beta x^{-1/6})^r$$

$$\text{For term independent of 'x' } r = 4 \left[\because \frac{10-r}{9} - \frac{r}{6} = 0 \right]$$

$$\therefore \text{Term independent of } x = {}^{10}C_r \alpha^6 \beta^4$$

$$\text{Also } \alpha^3 + \beta^2 = 4$$

By AM - GM inequality

$$\frac{\alpha^3 + \beta^2}{2} \geq (\alpha^3 \beta^2)^{1/2}$$

$$\Rightarrow (2)^2 \geq \alpha^3 \beta^2 \Rightarrow \alpha^6 \beta^4 \leq 16$$

$$\therefore 10k = {}^{10}C_4 \cdot 16 \Rightarrow k = 336$$

$$12. (c) \frac{{}^{36}C_{r+1}}{{}^{35}C_r} = \frac{6}{k^2 - 3} \Rightarrow \frac{36}{r+1} \cdot \frac{{}^{35}C_r}{{}^{35}C_r} = \frac{6}{k^2 - 3}$$

$$\left[\because {}^nC_r = \frac{n}{r} {}^{n-1}C_{r-1} \right]$$

$$k^2 - 3 = \frac{r+1}{6} \Rightarrow k^2 = 3 + \frac{r+1}{6}$$

r can be 5, 35

for $r = 5, k = \pm 2$

$r = 35, k = \pm 3$

Hence, numbers of order pair = 4

$$13. (b) \frac{(1+x)^{10} \left[1 - \left(\frac{x}{1+x} \right)^{11} \right]}{\left(1 - \frac{x}{1+x} \right)} = \text{Sum of G.P.}$$

$$= \frac{(1+x)^{10} [(1+x)^{11} - x^{11}]}{(1+x)^{11} \times \frac{1}{(1+x)}} = (1+x)^{11} - x^{11}$$

Coefficient of x^7 is ${}^{11}C_7 = {}^{11}C_4 = 330$

$$14. (b) (1+x)(1-x)^{10} (1+x+x^2)^9$$

$$= (1-x^2)(1-x^3)^9 \quad [\because 1-x^3 = (1-x)(1+x+x^2)]$$

Coeff. of x^{18}

$${}^9C_6 = 84$$

$$15. (b) \sum_{r=1}^{26} \frac{1}{(2r-1)!(51-(2r-1))!} = \sum_{r=1}^{26} \frac{{}^{51}C_{(2r-1)}}{51!}$$

$$= \frac{1}{51!} \{ {}^{51}C_1 + {}^{51}C_3 + \dots + {}^{51}C_{51} \} = \frac{1}{51!} (2^{50})$$

$$16. (c) S = 0.({}^{30}C_0)^2 + 1.({}^{30}C_1)^2 + 2.({}^{30}C_2)^2 + \dots + 30.({}^{30}C_{30})^2 \dots (i)$$

$$S = 30.({}^{30}C_0)^2 + 29.({}^{30}C_1)^2 + 28.({}^{30}C_2)^2 + \dots + 0.({}^{30}C_{30})^2 \dots (ii)$$

On adding both equations (i) and (ii), we get

$$2S = 30.({}^{30}C_0^2 + {}^{30}C_1^2 + \dots + {}^{30}C_{30}^2)$$

$$S = 15.{}^{60}C_{30} = 15 \cdot \frac{60!}{(30!)^2} \left[\because nC_0^2 + nC_1^2 + nC_n^2 = 2^n C_n \right]$$

$$\text{So, } \frac{15 \cdot 60!}{(30!)^2} = \frac{\alpha \cdot 60!}{(30!)^2}$$

$$\Rightarrow \alpha = 15$$

$$17. (b) \sum_{r=0}^n \frac{{}^nC_r}{r+1} = \frac{1}{n+1} \sum_{r=0}^n {}^{n+1}C_{r+1}$$

$$= \frac{1}{n+1} (2^{n+1} - 1) = \frac{1023}{10} \quad \left[\because {}^{n+1}C_{r+1} = \frac{n+1}{r+1} {}^nC_r \right]$$

By cooperating

$$\Rightarrow n+1 = 10 \Rightarrow n = 90$$

$$18. (a) \sum_{k=1}^{31} ({}^{31}C_k)({}^{31}C_{k-1}) = \sum_{k=1}^{31} ({}^{31}C_{31-k})({}^{31}C_{k-1}) = {}^{62}C_{30}$$

$$\text{and } \sum_{k=1}^{30} ({}^{30}C_k)({}^{30}C_{k-1}) = \sum_{k=1}^{30} ({}^{30}C_{30-k})({}^{30}C_{k-1}) = {}^{60}C_{29}$$

$$\text{Now, } {}^{62}C_{30} - {}^{60}C_{29} = \frac{62!}{30!32!} - \frac{60!}{29!31!}$$

$$= \frac{60!}{29!31!} \left\{ \frac{62 \cdot 61}{30 \cdot 32} - 1 \right\}$$

$$= \frac{60!}{30!31!} \left\{ \frac{2822}{32} \right\}$$

$$\therefore 16\alpha = 16 \times \frac{2822}{32} = 1411$$

$$19. (d) \left(3x^3 - 2x^2 + \frac{5}{x^5} \right)^{10}$$

$$\text{General term } 10! \frac{(3x^3)^\alpha \cdot (-2x^2)^\beta \cdot (5x^{-5})^\gamma}{\alpha! \beta! \gamma!}$$

$$= 10! \frac{3^\alpha \cdot (-2)^\beta \cdot 5^\gamma \cdot x^{3\alpha+2\beta-5\gamma}}{\alpha! \beta! \gamma!}$$

$$\text{Now for constant term } \Rightarrow 3\alpha + 3\beta - 5\gamma = 0 \dots (i)$$

$$\alpha + \beta + \gamma = 10$$

From equation (i) & (ii)

$$3\alpha + 2(10 - \alpha - \gamma) = 5\gamma$$

$$\alpha + 20 = 7\gamma$$

$$\alpha = 1, \gamma = 3, \beta = 6$$

Constant term $10! \cdot \frac{3^1 \cdot (-2)^6 \cdot 5^3}{3!6!} = 2^9 \cdot 3^2 \cdot 5^4 \cdot 7^1$

20. (d) $T_{r+1} = {}^{10}C_r (tx^{1/5})^{(10-r)} \left[\frac{(1-x)^{1/10}}{t} \right]^r$

$$= {}^{10}C_r t^{(10-2r)} \times x^{\frac{10-r}{5}} \times (1-x)^{\frac{r}{10}}$$

$$\Rightarrow 10 - 2r = 0 \Rightarrow r = 5$$

$$T_6 = {}^{10}C_5 \sqrt{1-x} \cdot x$$

$$\frac{dT_6}{dx} = {}^{10}C_5 \left[\sqrt{1-x} - \frac{x}{2\sqrt{1-x}} \right] = 0$$

$$= 1 - x = x/2 \Rightarrow 3x = 2$$

$$\Rightarrow x = 2/3$$

$$(T_6)_{\max} = \frac{10!}{5!5!} \times \frac{2}{3\sqrt{3}}$$

21. [29] $(21+2)^{200} + (21-2)^{200}$

$$\Rightarrow 2 \left[{}^{100}C_0 21^{200} + {}^{200}C_2 21^{198} \cdot 2^2 + \dots + {}^{200}C_{198} 21^2 \cdot 2^{198} + 2^{200} \right]$$

$$\Rightarrow 2 \left[49I_1 + 2^{200} \right] = 49I_1 + 2^{201}$$

Now,

$$2^{201} = (8)^{67} = (1+7)^{67} = 49I_2 + {}^{67}C_0 + {}^{67}C_1 \cdot 7$$

$$= 49I_2 + 470 = 49I_2 + 49 \times 9 + 29$$

$\therefore$ Remainder is 29

22. [171] The number of integral term in the expression of

$$\left(\frac{1}{3^2} + \frac{1}{5^4} \right)^{680} \text{ is equal to}$$

$$\text{General term} = {}^{680}C_r \left(\frac{1}{3^2} \right)^{680-r} \left(\frac{1}{5^4} \right)^r$$

$$= {}^{680}C_r 3^{\frac{680-r}{2}} 5^{\frac{r}{5}}$$

For integral terms $r = 4k$

So, $r = 0, 4, 8, 12, \dots, 680$

Now, number of terms in the series

$$680 = 0 + (n-1)4 \Rightarrow n = 171$$

23. [1] $T_{r+1} = {}^{22}C_r \cdot \left(x^{\frac{2}{3}} \right)^{22-r} \cdot (\alpha)^r \cdot x^{-3r}$

$$= {}^{22}C_r \cdot x^{\frac{44}{3} - \frac{2r}{3} - 3r} (\alpha)^r$$

for constant term

$$\frac{44}{3} - \frac{11r}{3} = 0 \quad [\text{The term without } x]$$

$$\Rightarrow r = 4$$

$$\text{So, } {}^{22}C_4 \cdot \alpha^4 = 7315$$

$$\Rightarrow \frac{22 \times 21 \times 20 \times 19}{24} \cdot \alpha^4 = 7315$$

$$\Rightarrow \alpha = 1$$

24. [84] $T_5 = {}^nC_4 \left(2^{\frac{1}{4}} \right)^{n-4} \cdot \left(\left(\frac{1}{3} \right)^{\frac{1}{4}} \right)^4 = {}^nC_4 \cdot 2^{\frac{n-4}{4}} \cdot \frac{1}{3}$

$$T_{n-3} = {}^nC_{n-4} \left(2^{\frac{1}{4}} \right)^4 \cdot \left(\left(\frac{1}{3} \right)^{\frac{1}{4}} \right)^{n-4} = {}^nC_{n-4} \cdot 2 \cdot \left(\frac{1}{3} \right)^{\frac{n-4}{4}}$$

$$\frac{T_5}{T_{n-3}} = \frac{2^{(n-8)/4}}{\left(\frac{1}{3} \right)^{(n-8)/4}} = 6^{(n-8)/4} = 6^{\frac{1}{4}} \Rightarrow n = 9$$

$$T_6 = {}^9C_5 \left(2^{\frac{1}{4}} \right)^4 \cdot \left(\left(\frac{1}{3} \right)^{\frac{1}{4}} \right)^5 = {}^9C_5 \cdot 2 \cdot \frac{1}{3^{5/4}} = \frac{2}{3} \cdot \frac{1}{3^{\frac{1}{4}}} \cdot \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = \frac{84}{3^{\frac{1}{4}}}$$

$$\Rightarrow \alpha = 84$$

25. [210] $\left(\frac{x+1}{x^{2/3} - x^{1/3} + 1} - \frac{x-1}{x-x^{1/2}} \right)^{10}$

$$= \frac{(x^{1/3})^3 + 1^3}{x^{2/3} - x^{1/3} + 1} - \frac{(\sqrt{x})^2 - 1^2}{\sqrt{x}(\sqrt{x}-1)}$$

$$= \frac{(x^{1/3} + 1)^3 (x^{2/3} + 1 - x^{1/3})}{x^{2/3} - x^{1/3} + 1} - \frac{\sqrt{x} + 1}{\sqrt{x}}$$

$$= \left((x^{1/3} + 1) - (1 + x^{-1/2}) \right)^{10}$$

$$= (x^{1/3} - x^{-1/2})^{10}$$

$$\text{General term } T_r = {}^{10}C_r (-1)^r x^{\frac{10-r}{3}} \cdot x^{-\frac{r}{2}}$$

$$= {}^{10}C_r (-1)^r x^{\frac{20-5r}{6}}$$

For independent of $x, 20 - 5r = 0 \Rightarrow r = 4$

$$\text{Coefficient} = {}^{10}C_4 = 210$$

$$26.[315] \quad a^{10}b^{10} \left(\frac{1}{b} + \frac{2}{a} + 4 \right)^{10}$$

$$a^{10}b^{10} \frac{10! \left(\frac{1}{b} \right)^{r_1} \left(\frac{2}{a} \right)^{r_2} \cdot 4^{10-r_1-r_2}}{r_1! r_2! (10-r_1-r_2)!}$$

$$r_1 = 2, r_2 = 3$$

$$a^7 b^8 \text{ is } \frac{10! 2^3 \cdot 4^{10-2-3}}{2! 3! 5!} = \frac{2^{13} \cdot 10!}{2! 3! 5!} = 2^{16} \cdot 315$$

$$K = 315$$

$$27. [8] \quad (2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$$

$$\text{General term} = \frac{10!}{r_1! r_2! r_3!} (2x^2)^{r_1} (3x)^{r_2} (4)^{r_3}$$

As a^7 = coeff of x^7

$$2r_1 + r_2 = 7 \text{ and } r_1 + r_2 + r_3 = 10$$

Possibilities are

r_1	r_2	r_3
0	7	3
1	5	4
2	3	5
3	1	6

$$a_7 = \frac{10! 3^7 4^3}{7! 3!} + \frac{10! (2)(3)^5 (4)^4}{5! 4!} + \frac{10! (2)^2 (3)^3 (4)^5}{2! 3! 5!} + \frac{10! (2)^3 (3)(4)^6}{3! 6!}$$

$$a_{13} = \text{Coeff of } x^{13}$$

$$2r_1 + r_2 = 13 \text{ and } r_1 + r_2 + r_3 = 10$$

Possibilities are

r_1	r_2	r_3
3	7	0
4	5	1
5	3	2
6	1	3

$$a_{13} = \frac{10! (2^3)(3^7)}{3! 7!} + \frac{10! (2^4)(3^5)(4)}{4! 5!}$$

$$+ \frac{10! (2^5)(3^3)(4^2)}{5! 3! 2!} + \frac{10! (2^6)(3)(4^3)}{6! 1! 3!}$$

$$\text{Clearly } \frac{a_7}{a_{13}} = 2^3 = 8$$

$$28. [51]$$

$$\begin{aligned} \sum_{r=0}^{25} (4r+1)^{25} C_r &= 4 \sum_{r=0}^{25} r^{25} C_r + \sum_{r=0}^{25} {}^{25}C_r \\ &= 4 \sum_{r=1}^{25} r \times \frac{25}{r} {}^{25}C_{r-1} + 2^{25} = 100 \sum_{r=1}^{25} {}^{25}C_{r-1} + 2^{25} \\ &= 100 \cdot 2^{24} + 2^{25} = 2^{25} (50 + 1) = 51 \cdot 2^{25} \end{aligned}$$

$$\text{So } k = 51$$

$$29. [102]$$

$$\begin{aligned} & {}^{40}C_0 + {}^{41}C_1 + {}^{42}C_2 + \dots + {}^{59}C_{19} + {}^{60}C_{20} \\ &= ({}^{41}C_0 + {}^{41}C_1) + {}^{42}C_2 + \dots + {}^{59}C_{19} + {}^{60}C_{20} \\ &= ({}^{42}C_1 + {}^{42}C_1) + \dots + {}^{59}C_{19} + {}^{60}C_{20} \\ &= {}^{43}C_2 + {}^{43}C_3 + \dots + {}^{59}C_{19} + {}^{60}C_{20} \end{aligned}$$

$$\dots \dots$$

$$= {}^{60}C_{19} + {}^{60}C_{20} = {}^{61}C_{20} = \frac{61}{41} {}^{60}C_{20}$$

$$\therefore m = 61 \text{ and } n = 41$$

$$\Rightarrow m + n = 102.$$

$$30. [49] \quad A_k = \sum_{i=0}^9 {}^9C_i \times {}^{12}C_{k-i} + \sum_{i=0}^8 {}^8C_i \times {}^{13}C_{k-i}$$

$$A_4 = \sum_{i=0}^9 {}^9C_i \times {}^{12}C_{4-i} + \sum_{i=0}^8 {}^8C_i \times {}^{13}C_{4-i}$$

$$= \text{coefficient of } x^4 \text{ in } (1+x)^{21} \text{ coefficient of } x^4 \text{ in } (1+x)^{21} \\ = {}^{21}C_4 + {}^{21}C_4 = 2 \cdot {}^{21}C_4$$

$$A_3 = \text{coefficient of } x^3 \text{ in } (1+x)^{21} + \text{coefficient of } x^3 \text{ in } (1+x)^{21}$$

$$= {}^{21}C_3 + {}^{21}C_3 = 2 \cdot ({}^{21}C_3)$$

$$A_4 - A_3 = 2 [{}^{21}C_4 - {}^{21}C_3] = 190p$$

$$\Rightarrow 2[5985 - 1330] = 190p$$

$$p = 49$$

